National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Faculty of Computing

Department of Computing

Laboratory Manual

CS-250 : Data Structure and Algorithms

Class: BESE 16AB • Semester: Fall 2026

Experiment No. 6

AVL Trees

Date	16 October 2026
Time	09:00 AM–11:50 AM & 2:30 PM–5:00 PM
Instructor(s)	Dr. Faisal Shafait
Lab Engineer	Hamza Fayyaz
Mapped CLO	**CLO2 : Solve a given real time problem by applying appropriate data structure and algorithm.** **CLO4 : Demonstrate commitment to professional ethics by following engineering norms applicable to the software industry.**:

1 Introduction

Binary Search Trees (BSTs) support searching, insertion, and data management, but sorted or nearly sorted insertions may produce a skewed tree of height $O(n)$, degrading operations to $O(n)$. AVL trees are self-balancing BSTs. Each node's balance factor must be -1, 0, or $+1$. Rotations restore balance after insertions or deletions, maintaining height $O(\log n)$ and logarithmic search, insertion, and deletion.

2 Objectives

- Understand height-balanced binary search trees and identify critical imbalance.
- Differentiate LL, RR, LR, and RL rotations.
- Implement AVL insertion and the rotations required to maintain balance.

3 Tools and Software Requirements

Software: Visual Studio Code

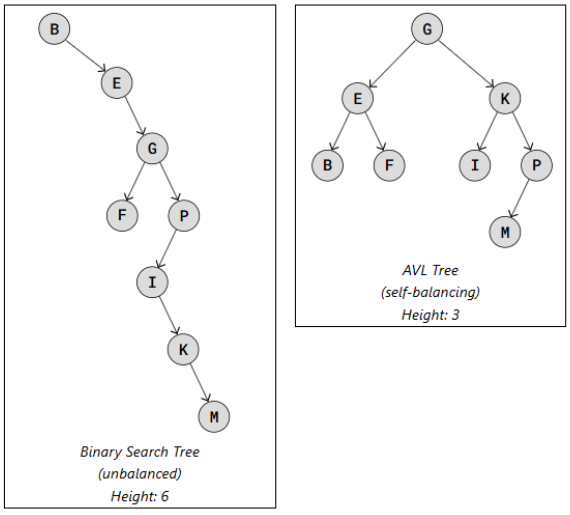
Helping Material: Lecture slides, textbook

4 Description

An AVL tree maintains the BST property while keeping the heights of its left and right subtrees balanced. The four imbalance corrections are:

- LL: right rotation
- RR: left rotation
- LR: left-right rotation
- RL: right-left rotation

The diagram below, taken from the supplied manual, contrasts an unbalanced BST with an AVL tree.



Balance Factor

The balance factor is the difference between the left and right subtree heights:

$$BF(v) = h(v_{\text{left}}) - h(v_{\text{right}})$$

For an AVL tree, $BF(v) \in \{-1, 0, +1\}$ for every node.

5 Lab Tasks

Task 1: AVL Tree Insertion and Rotations

Write a C++ program that:

1. Creates an AVL tree.
2. Inserts *N* numbers provided by the user.
3. Checks for imbalance after each insertion.
4. Applies the required LL, RR, LR, or RL rotation.
5. Displays the updated tree using in-order traversal.

Task 2: Height and Balance Factor of Each Node

Extend Task 1 with functions to calculate each node's height and balance factor. Display the results in the following format:

Listing 1. *Example Output*

```
Node: 25 | Height: 2 | Balance Factor: -1
Node: 20 | Height: 1 | Balance Factor: 1
Node: 30 | Height: 2 | Balance Factor: 0
```

6 Deliverables

Complete the solution sections and submit a single Word document on LMS before the deadline. Demonstrate your implementation during the lab. Use proper indentation and comments; missing comments or poor indentation may result in deduction of marks.

7 Important Note

Practice object-oriented programming in C++ and comment your code appropriately. Inappropriate or missing comments may result in deduction of marks.

8 Lab Rubrics

Assessment	Doesn't meet Expectation (1–2)	Meets Expectation (3–4)	Exceeds Expectation (5)	Marks
Software Problem Realization (CLO2)	The Student is unable to understand and outline the problem and doesn't use the relevant method to solve it.	The student requires some guidance to completely comprehend the problem and to differentiate the data structure and algorithm comprehensively.	The student fully understands the given problem, is able to analyze the relevant method to solve it, and develops a detailed program flow.	
Ethics and Adherence to Laboratory Safety Rules (CLO4)	The student does not behave according to the professional ethics by following ethical norms applicable to the software industry such as acknowledgement while using publicly available data/code. Disturbs the lab environment, doesn't take care of safety measures, and/or isn't punctual.	The student partially demonstrate their commitment to professional ethics by following ethical norms applicable to the software industry such as referencing and acknowledgement while using publicly available data/code. Exhibits better behavior, works by taking into account the safety measures, and is punctual.	The student clearly express the commitment to professional ethics by following ethical norms applicable to the software industry such as referencing and acknowledgement while using publicly available data/code. Encourages others to maintain lab decorum, and alerts them to follow safety measures.	